Domain Transform Model Driven by Deep Learning for Anti-Noise Hyperspectral and Multispectral Image Fusion

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***Abstract*— While fusion of hyperspectral images (HSIs) with low spatial resolution and multispectral images (MSIs) with high spatial resolution has achieved significant success, high-quality fusion between noisy images has always been challenging. In this article, we propose a domain transform model driven by deep learning for anti-noise hyperspectral and multispectral image fusion (DTAFN). This marks the first time that wavelet decomposition theory is combined with deep learning for noise reduction in hyperspectral and MSI fusion. DTAFN initially decomposes hyperspectral and MSIs into frequency components and constructs a novel feature interaction fusion module (FIFM). This module, while using MSIs to guide the removal of noise from HSIs, also achieves the fusion of spatial and spectral information. Furthermore, it maps the fused features to a lower dimensional subspace to enhance computational efficiency. Additionally, we introduce a spatial–spectral self-attention mechanism to optimize the reconstructed frequency components using the subspace features. In the end, the wavelet inverse transform is used to reconstruct the clean fused image. It is worth noting that the extraction of the subspace is considered a process of nonlinear low-rank component extraction, which, to a certain extent, suppresses noise signals. Numerous experiments of mixed noise image fusion are carried out, and the experimental results show that DTAFN can obtain high-quality fusion results, is robust, and superior to the state-of-the-art methods.**

***Index Terms*— Anti-noise, domain transform, fusion, hyperspectral, multispectral.**

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I. Introduction

HYPERSPECTRAL satellites usually cover the 400–2500-nm electromagnetic spectrum and can obtain more than 100 channels [1], [2]. Hyperspectral image (HSI) is usually used for fine surface monitoring. However, the narrow channel (10–20 nm) has limited absorption for surface reflected energy and is easily affected by weather and equipment factors, which makes the HSI usually polluted by serious noise, has low spatial resolution, and is seriously limited for fine surface monitoring [3]. On the contrary, multispectral satellites typically cover the 400–900-nm electromagnetic spectrum, and the number of channels is less than 10 [4]. The wider channels enable the multispectral image (MSI) to absorb more reflected energy, which is not easy to be polluted by noise and has high spatial resolution. However, low spectral resolution makes it impossible to use for fine surface monitoring [5], [6].

Image fusion methods aim to combine spectral information of HSI and spatial information of MSI by using certain rules for the enhance of spatial resolution of HSI [7], [8]. These methods can be roughly divided into component substitution (CS) methods [9], multiresolution analysis (MRA) methods [10], matrix decomposition methods, Bayesian methods, and deep learning methods [11], [12]. CS methods first perform forward transformation to extract the intensity components of HSI. Then, MSI is used to replace the intensity component of HSI, and finally, inverse transformation is performed to achieve image fusion [13]. Typical methods include principal component analysis (PCA) [14], adaptive Gram–Schmidt (GSA) [15], and band-dependent spatial detail (BDSD) [16]. Due to the replacement of intensity component, the fusion results of CS have clear spatial details. However, there is an unequal gray distribution between the intensity component of HSI and MSI, which causes serious spectral distortion [17]. MRA methods first decompose the image at multiscale and calculate the spatial difference information between HSI and MSI at different scales and finally inject the

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spatial difference information into the HSI to achieve fusion [18]. Specific methods include modulation transfer function-generalized Laplacian pyramid (MTF-GLP) [19], smoothing filter-based intensity modulation (SFIM) [20], and "a Ì trous" wavelet transform (ATWT) [21]. MRA methods do not change the spectral characteristics of the original HSI, so it has high spectral fidelity. However, some spatial information from MSI has not been injected into HSI, and the fused image has spatial distortion [22]. Matrix decomposition methods introduce the theory of mixed pixel decomposition, sparse representation, and tensor decomposition into image fusion, believe that the image can be decomposed into spectral matrix and spatial matrix, and image fusion is realized by restructuring the spectral matrix of HSI and spatial matrix of MSI by using constraint rules [23]. Typical methods include coupled nonnegative matrix factorization (CNMF) based on mixed pixel decomposition theory [24], and Li et al. [25] and Dian et al. [26], [27] first proposed a tensor representation-based method, which obtained superior performance and became a new branch in HSI and MSI fusion. Matrix decomposition methods obtain the optimal solution through iterative optimization and have good spectral and spatial fidelity. However, these methods have many parameters, low robustness, and computational efficiency [28]. Bayesian methods take the original HSI and MSI as a prior probability and achieve image fusion by obtaining the maximum posterior probability [11]. Fast fusion based on Sylvester equation (FUSE) is widely used in image fusion as typical Bayesian methods [29]. However, when the ground objects in the image are complex, Bayesian methods cannot obtain high-quality fused images [1]. In recent years, deep learning methods have been widely used in image processing and have achieved comparative advantages [30], [31], [32]. He et al. [33] proposed a convolutional neural networks (CNNs)-based HSI fusion model (ARHS-CNN) in 2021, which can train a CNN model in a limited number of samples to convert HSI to arbitrary spatial resolution. Ma et al. [34] proposed an unsupervised pan-sharpening framework (Pan-GAN) based on generative adversarial network. Pan-GAN established an adversarial game between the generator and the spectral discriminator and the spatial discriminator, respectively, and achieved comparative fusion results by continuously optimizing the generator. Bandara and Patel [35] proposed a transformer network based on attention mechanism for hyperspectral pan-sharpening (HyperTransformer). From the perspective imaging model, Dian et al. [9] proposed a seminal work for high-resolution hyperspectral acquisition, which accurately estimated the imaging model for HSI and MSI fusion and achieved excellent fusion performance with high computational efficiency. Deep learning methods solve fusion parameters nonlinearly and obtain high spectral and spatial fidelity. However, numerous training data have brought low computational efficiency [3].

It is worth noting that due to the influence of weather, equipment, and other factors, HSI is seriously polluted by noise, especially mixed noise [36]. Although the existing methods can yield promising results under noise-free conditions, many of these methods often overlook the realistic scenario of noise in HSI. When noise is present, the spatial structures and internal spectral relations of the HSI are compromised during the fusion process. Moreover, the structural artifacts induced by noise are even more challenging to remove with conventional methods, mainly due to the lack of consideration for different noise distribution properties. Although the data quality can be improved by step processing (fusion after noise removal or noise removal after fusion), the accumulation of errors between models has caused serious spectral and spatial distortion. Fig. 1 shows the noisy HSI fusion results of different fusion methods (CNMF, GSA, and MTF-GLP) combined with different denoising methods (low rank matrix recovery (LRMR) [37], block matching 4D (BM4D) [38], and local low rank total variation (LLRT) [39]). Considering the low-rank properties of HSI and the nonlinear frequency distribution characteristics of mixed noisy signals, we propose a domain transform model driven by deep learning for anti-noise hyperspectral and multispectral image fusion (DTAFN). To better describe the distribution characteristics of noisy signals, DTAFN first converts HSI and MSI into frequency domains by using discrete wavelet transform (DWT). On the one hand, the multifrequency components of the image are obtained, which effectively separates some noisy signals. On the other hand, frequency conversion reduces the size of the original image and improves computational efficiency. Afterward, the frequency components of HSI and MSI are fused using the feature interaction fusion module (FIFM). FIFM employs a feature interaction fusion approach, which, on the one hand, utilizes MSI to guide the extraction of noise signals from HSI. On the other hand, it incorporates a Conv-LSTM network to extract subspace features from the fused frequency components. This process effectively diminishes redundancy among feature layers, isolates noisy signals, and eliminates them through the use of the forgetting gate. Consequently, the extracted subspace features exhibit low-rank properties. After that, a subspace back projection module (SBPM) considering local and global characteristics is proposed to reconstruct high spatial resolution frequency components without noise. Finally, DTAFN performs inverse wavelet transform (IWT) of frequency components to achieve image fusion and refine the fusion results by using convolutional layer. The main contributions of this article are as follows.

1) We propose DTAFN for the fusion of anti-noise HSI and MSI. DTAFN fully considers the frequency distribution characteristics of noise and the nonlinear problems existing in image fusion and combines traditional DWT and deep learning framework for the first time to effectively suppress noisy signals while minimizing spatial–spectral distortion, effectively improving the fusion performance of noisy HSI and MSI.
2) In DTAFN, we have constructed a novel FIFM for feature fusion. On the one hand, it uses MSI as an auxiliary image to denoise HSI. On the other hand, it effectively fuses spatial–spectral features. Additionally, the use of Conv-LSTM significantly enhances computational efficiency. We have also designed a spatial–spectral self-attention network to propose SBPM, which effectively optimizes the fused features and accurately

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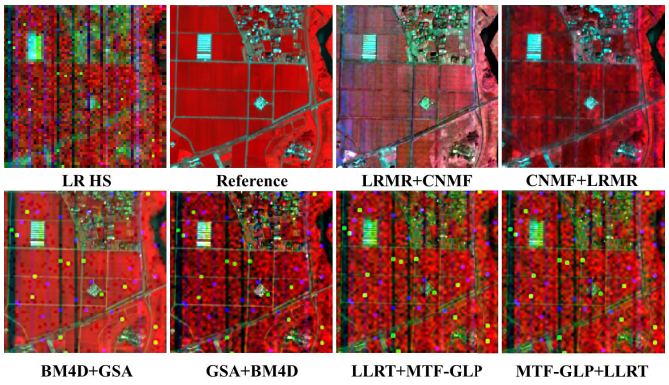


Fig. 1. Fusion results of fusion after noise removal and noise removal after fusion.

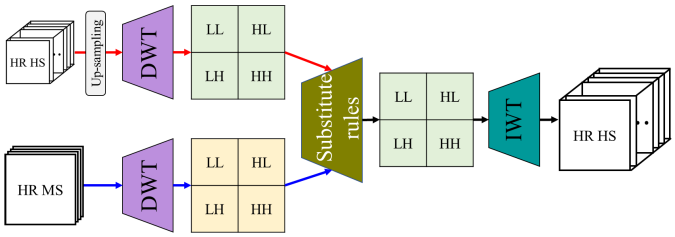


Fig. 2. DWT fusion method based on CS.

reconstructs the clean frequency components with high spatial resolution.

3) DTAFN effectively suppresses noise in different datasets with mixed noise, accurately reconstructs HSI with high spatial resolution, and outperforms nine state-of-the-art methods in noise suppression, computational efficiency, spectral, and spatial fidelity.

The structure of this article is as follows. Section II introduces the methodology, Section III describes the experimental setup and results, and the conclusion is given in Section IV.

II. Methodology

A. DWT Fusion Model

Jorge et al. first proposed the pan-sharpening method based on DWT for MSI and PAN fusion in 1999 [40]. DWT can achieve image fusion through CS and MRA. DWT based on CS first performs wavelet decomposition on MSI and panchromatic image (PAN), replaces high-frequency components of MSI with high-frequency components of PAN, and finally realizes image fusion through IWT (see Fig. 2). DWT based on MRA only performs wavelet decomposition on PAN and obtains fused images by injecting high-frequency components into MSI (see Fig. 3). In recent years, Chen et al. [41] have proposed an extended framework of the pan-sharpening methods, which groups images by calculating the correlation between bands of HSI and MSI images and uses pan-sharpening methods to fuse these grouped images. Vivone et al. [42] synthesized PAN using MSI image and fused the synthesized PAN with HSI using pan-sharpening methods. Therefore, DWT has also been used for HSI and MSI fusion in recent years.

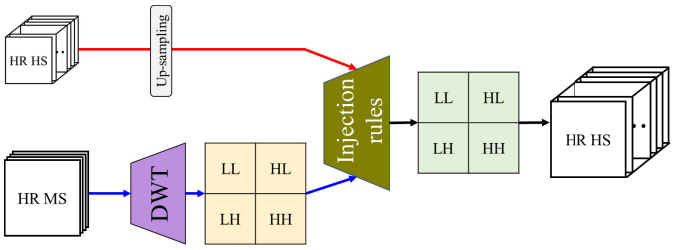


Fig. 3. DWT fusion method based on MRA.

However, DWT based on CS has serious spectral distortion, which is caused by poor substitution rules that change the spectral properties of the original image. The injection rule of DWT based on MRA has limited improvement in spatial resolution, and numerous spatial details are lost, which causes serious spatial distortion.

B. DTAFN

DWT is used to perform domain transformation in image processing. On the one hand, frequency features can be used to describe the distribution characteristics of noise signals more accurately. On the other hand, multifrequency information of the image can be obtained to extract more abundant features, and downscaling speeds up computational efficiency. However, DWT has a serious distortion of spectral and spatial information in image fusion, and an accurate fusion rule is needed. To solve the above problems and improve fusion performance, we propose DTAFN. DTAFN first upsamples noisy HSI, and DWT is performed for the upsampled noisy HSI and MSI. Then, the multifrequency features of noisy HSI and MSI are grouped accordingly. Considering the nonlinear differences caused by different resolution and gray values between images and the necessity of low rank component extraction for noise image restoration, Conv-LSTM is used to extract subspaces of grouping features. After that, a subspace back projection network is proposed, and the clean frequency subspaces are used to calculate the optimal multifrequency components of the fused image. Finally, the obtained multifrequency components are used for IWT to realize the fusion of noisy HSI and MSI. Indeed, it is worth noting that our method aims to separate and remove noise signals through nonlinear regression. In the signal separation phase, we utilize the DWT to describe the frequency distribution characteristics of noise and clean signals, which allows us to maximize the separation of different signal components. In the signal removal phase, we leverage the forget gate operation in convolutional long short-term memory (Conv-LSTM) to discard signal components that are not relevant for reconstructing the ground truth. Due to supervised learning, all the parameters in the entire network training process are adaptively adjusted.

1) DWT Forward Transform: HSI is seriously affected by noise signals. Considering that frequency features can more accurately describe the distribution properties of noise signals, DWT is used to perform domain transformations on noisy HSI and MSI. DTAFN introduces DWT redefined by tensor operations proposed by Liu et al., [40] which can fit deep learning networks and effectively perform gradient feedback.

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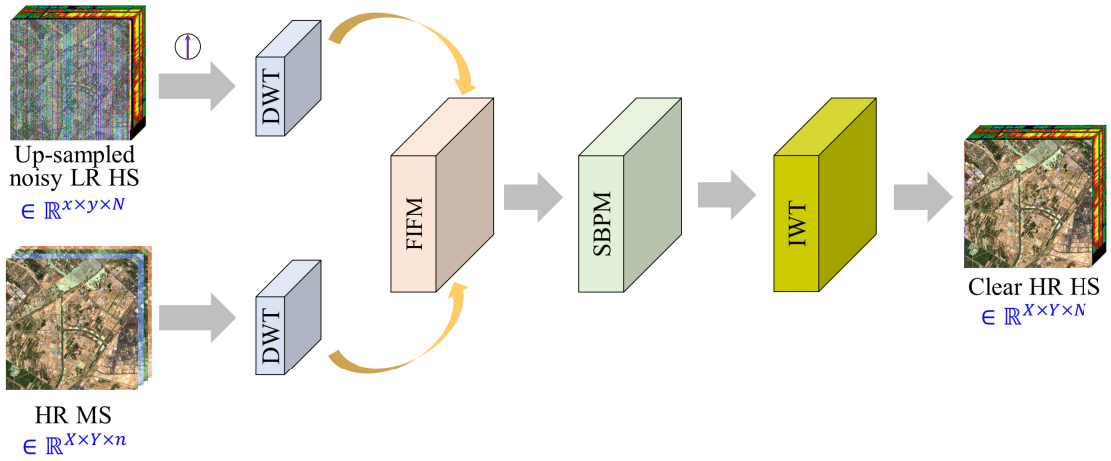


Fig. 4. DTAFN.

The DWT consists of a low-pass filter ψ_{ll} and three high-pass filters ψ_{hh}, ψ_{hl}, and ψ_{lh}, and these filters have a fixed parameter of convolution step of 2. The convolution kernel is defined as follows:

$$\psi_{ll} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \tag{1}$$

$$\psi_{hh} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad \psi_{hl} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\psi_{lh} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}. \tag{2}$$

ψ_{ll}, ψ_{hh}, ψ_{hl}, and ψ_{lh} are orthogonal to each other and form an invertible matrix. Thus, the different components of the decomposed image can be expressed as follows:

$$\mathbf{Z}_{ll} = S \cdot (\mathbf{Z} \otimes \psi_{ll}) \tag{3}$$

$$\mathbf{Z}_{hh} = S \cdot (\mathbf{Z} \otimes \psi_{hh})$$

$$\mathbf{Z}_{hl} = S \cdot (\mathbf{Z} \otimes \psi_{hl})$$

$$\mathbf{Z}_{lh} = S \cdot (\mathbf{Z} \otimes \psi_{lh}) \tag{4}$$

where $\mathbf{Z}$ is the decomposed image, S is the downsampling operation with a factor of 2, and $\otimes$ is the convolution operation. According to Haar transformation theory, DWT can be further expressed as follows:

$$\begin{aligned}
\mathbf{Z}_{ll}(m,n) &= \mathbf{Z}(2m-1,2n-1) + \mathbf{Z}(2m-1,2n) \\
&\quad + \mathbf{Z}(2m,2n-1) + \mathbf{Z}(2m,2n) \\
\mathbf{Z}_{hh}(m,n) &= \mathbf{Z}(2m-1,2n-1) - \mathbf{Z}(2m-1,2n) \\
&\quad - \mathbf{Z}(2m,2n-1) + \mathbf{Z}(2m,2n) \\
\mathbf{Z}_{hl}(m,n) &= -\mathbf{Z}(2m-1,2n-1) + \mathbf{Z}(2m-1,2n) \\
&\quad - \mathbf{Z}(2m,2n-1) + \mathbf{Z}(2m,2n) \\
\mathbf{Z}_{lh}(m,n) &= -\mathbf{Z}(2m-1,2n-1) - \mathbf{Z}(2m-1,2n) \\
&\quad + \mathbf{Z}(2m,2n-1) + \mathbf{Z}(2m,2n)
\end{aligned} \tag{5}$$

where (m,n) represents the pixel position of $\mathbf{Z}_{ll}$, $\mathbf{Z}_{hh}$, $\mathbf{Z}_{hl}$, and $\mathbf{Z}_{lh}$.

Finally, HSI and MSI are decomposed as follows:

$$[\mathbf{H}_{ll}, \mathbf{H}_{hh}, \mathbf{H}_{hl}, \mathbf{H}_{lh}] = \mathrm{DWT}(R(\mathbf{HS})) \tag{6}$$

$$[\mathbf{M}_{ll}, \mathbf{M}_{hh}, \mathbf{M}_{hl}, \mathbf{M}_{lh}] = \mathrm{DWT}(\mathbf{MS}) \tag{7}$$

where R is the upsampling operation and DWT$(\cdot)$ is the wavelet decomposition operation.

2) Multifrequency Feature Fusion: To fuse the frequency components of HSI and MSI, we combined conv blocks with Conv-LSTM to create the FIFM. FIFM consists of two convolution branches that extract the frequency component features of HSI and MSI. The features obtained from these convolution branches are then input to the Conv-LSTM. After passing through Conv-LSTM, the fused features are fed back into the convolution branches of HSI and MSI. FIFM effectively utilizes MSI to guide HSI in noise removal and obtains more precise fusion features through feature interaction. In the experiment, Conv-LSTM is used to extract the subspace of the fused frequency components [43], [44]. This choice is based on two primary reasons. First, as the network depth increases, CNN tends to introduce information redundancy between features, which can increase computational burden and make it challenging for the network to converge. Conv-LSTM helps to retain long-term information through cell state and calculate the difference information between features, thus reducing redundancy among hierarchical features. Second, Conv-LSTM uses input and forgetting gates to learn the dependencies between different feature layers. This allows it to eliminate invalid information from the cell state of the previous layer while retaining valid information for input to the next layer. In essence, Conv-LSTM can preserve clean features with low-rank properties and eliminate noisy signals with sparse properties (see Fig. 5). Specifically, the Conv block consists of three layers of convolution followed by ReLU activation, and the convolution kernel size is 3. The first Conv-LSTM consists of two cells, while the other Conv-LSTM models consist of three cells.

In Conv-LSTM [see Fig. 6], the forgetting gate determines which information needs to be discarded by transmitting the information from the previous hidden state and the current input information to the sigmoid function. The output value of the forgetting gate is between 0 and 1. These values close to 0 mean that the information should be discarded and close to 1 means that the information should be retained. Then, a new cell state is constructed by transmitting effective information to the input gate. After that, the previous hidden state, the

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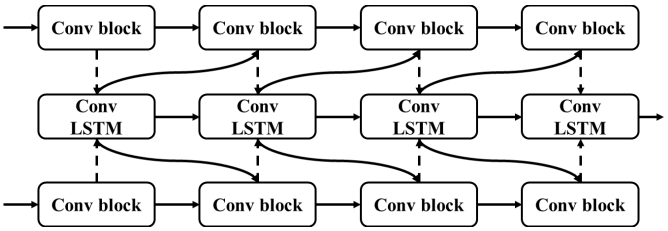


Fig. 5. Feature interaction fusion module.

current input data, and the current cell state are transferred to the sigmoid function, the new cell state is transferred to the tanh function, and the output of the tanh and the output of the sigmoid are multiplied to determine the information that the hidden state should carry. The final hidden state is used as the output of the current cell, and the new cell state and hidden state are transmitted to the next level. The specific formula is defined as follows:

$$f_t = \delta(w_f F_t + w_{hf} * H_{t-1} + w_{cf} \odot C_{t-1} + b_f) \tag{8}$$

$$i_t = \delta(w_i F_t + w_{hi} * H_{t-1} + w_{ci} \odot C_{t-1} + b_i) \tag{9}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(w_c * F_t + w_{hc} * H_{t-1} + b_c) \tag{10}$$

$$O_t = \delta(w_o * F_t + w_{ho} * H_{t-1} + w_{co} \odot C_t + b_o) \tag{11}$$

$$H_t = O_t \odot \tanh(C_t) \tag{12}$$

where δ is the sigmoid activation function, w is the weight, b is the bias, C is the cell state, t is the tth time, H is the input of the previous cell, F is the input of the current cell, i is the input gate, f is the forgetting gate, O is the output gate, $\odot$ is the multiplication operation, and $*$ is the convolution operation.

Finally, DTAFN input $[\mathbf{H}_{ll}, \mathbf{M}_{ll}]$, $[\mathbf{H}_{hh}, \mathbf{M}_{hh}]$, $[\mathbf{H}_{hl}, \mathbf{M}_{hl}]$, and $[\mathbf{H}_{lh}, \mathbf{M}_{lh}]$ into the subspace extraction block to obtain the subspaces $[\mathbf{F}_{ll}^{lr}, \mathbf{F}_{hh}^{lr}, \mathbf{F}_{hl}^{lr}, \mathbf{F}_{lh}^{lr}]$ of the fused frequency features, where $\mathbf{F}^{lr}$ represents the subspace.

For SBPM (see Fig. 7), DTAFN constructs a self-attention back projection network by considering the global and local correlation of the image, which effectively maps low dimensional features into high-dimensional spaces. SBPM includes spatial self-attention networks and spectral self-attention networks. For spectral self-attention networks, convolutional operation is performed on the subspace of fused frequency features to obtain three feature maps with different channel numbers. The specific formula is expressed as

$$u = \sigma(\mathcal{F}^{1\times1}(\sigma(\mathcal{F}^{3\times3}(\mathcal{T}_i)))) \tag{13}$$

$$v = \sigma(\mathcal{F}^{1\times1}(\mathcal{T}_i)) \tag{14}$$

$$\bar{u} = \sigma(\mathcal{F}^{1\times1}(\sigma(\mathcal{F}^{3\times3}(\mathcal{T}_i)))) \tag{15}$$

where $u \in \mathbb{R}^{c\times w\times k}$, $v \in \mathbb{R}^{c\times w\times K}$, and $\bar{u} \in \mathbb{R}^{c\times w\times k}$ are the convolution features, $\mathcal{F}^{3\times3}$ and $\mathcal{F}^{1\times1}$ are the convolution operation of 3×3 and 1×1, and σ is the ReLU activation function. After that, the dot product operation is performed on u and v, and softmax is used to calculate the spectral correlation matrix between u and v, and the correlation matrix and $\bar{u}$ are used to perform the dot product operation to obtain

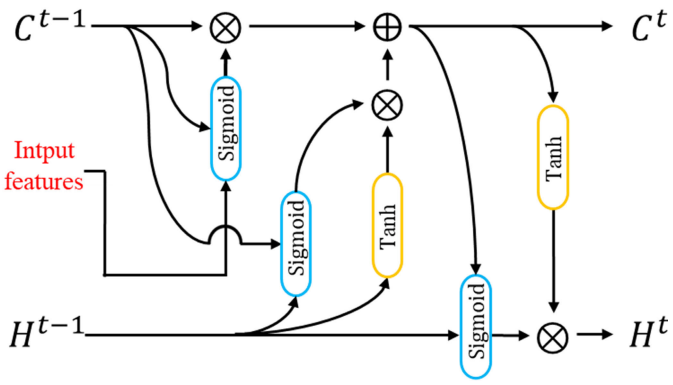


Fig. 6. Conv-LSTM cell.

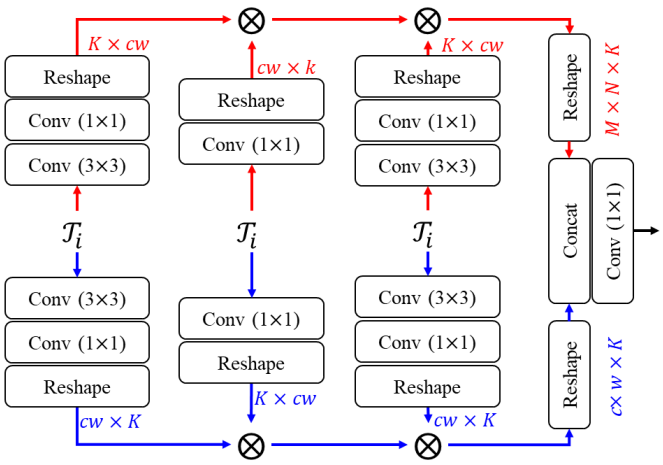


Fig. 7. Subspace back projection module.

the final attention characteristics

$$s = \text{softmax}(u \odot v) \tag{16}$$

$$\mathcal{P} = s^T \odot \bar{u} \tag{17}$$

where $s \in \mathbb{R}^{k\times K}$ is the correlation matrix, $\mathcal{P} \in \mathbb{R}^{c\times w\times K}$ is the spectral attention feature, and T is the transposition operation.

For spatial self-attention networks, the process is similar to spectral self-attention networks. However, in this case, the three feature components obtained after convolution have the same number of channels and spatial dimensions. They can be represented as $\hat{u} \in \mathbb{R}^{c\times w\times K}$, $\hat{v} \in \mathbb{R}^{c\times w\times K}$, and $\overline{\overline{u}} \in \mathbb{R}^{c\times w\times K}$. Furthermore, $\hat{u}$ and $\hat{v}$ are subject to dot product and are normalized through softmax to obtain a spatial correlation matrix, denoted as $\hat{s} \in \mathbb{R}^{cw\times cw}$. This can be expressed as $\hat{s} = \text{softmax}(\hat{u} \odot \hat{v})$. Finally, the correlation matrix is used to obtain the ultimate spatial attention features $\hat{\mathcal{P}}$ through dot product with $\overline{\overline{u}}$, and it can be represented as $\hat{\mathcal{P}} = \hat{s} \odot \overline{\overline{u}}^T$. After obtaining the spatial–spectral attention features, these features are stacked and then subjected to convolution to obtain the final features $\mathbf{F}$

$$\mathbf{F} = \sigma(\mathcal{F}^{1\times1}(\text{cat}(\mathcal{P}, \hat{\mathcal{P}}))). \tag{18}$$

Finally, DTAFN performs subspace back-projection transformation on $\mathbf{F}_{ll}^{lr}, \mathbf{F}_{hh}^{lr}, \mathbf{F}_{hl}^{lr}$, and$\mathbf{F}_{lh}^{lr}$, and the expected multifrequency components are $\mathbf{F}_{ll}$, $\mathbf{F}_{hh}$, $\mathbf{F}_{hl}$, and $\mathbf{F}_{lh}$.

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3) IWT Inverse Transform: Due to the biorthogonal characteristics of DWT, IWT can effectively reconstruct images without any information loss by using frequency components. Therefore, after obtaining the optimal multifrequency components, DTAFN uses IWT to perform the inverse transformation on the frequency components, which can be expressed as

$$\mathbf{X} = \mathrm{IWT}(\mathbf{F}_{ll}, \mathbf{F}_{hh}, \mathbf{F}_{hl}, \mathbf{F}_{lh}) \tag{19}$$

where $\mathbf{X}$ is the fused image obtained after inverse transformation, and IWT can be further expressed as

$$\begin{aligned}
\mathbf{X}(2m-1, 2n-1) &= (\mathbf{F}_{ll}(m,n) + \mathbf{F}_{lh}(m,n) \\
&\quad - \mathbf{F}_{hl}(m,n) + \mathbf{F}_{hh}(m,n))/4 \\
\mathbf{X}(2m-1, 2n) &= (\mathbf{F}_{ll}(m,n) - \mathbf{F}_{lh}(m,n) \\
&\quad + \mathbf{F}_{hl}(m,n) - \mathbf{F}_{hh}(m,n))/4 \\
\mathbf{X}(2m, 2n-1) &= (\mathbf{F}_{ll}(m,n) + \mathbf{F}_{lh}(m,n) \\
&\quad - \mathbf{F}_{hl}(m,n) - \mathbf{F}_{hh}(m,n))/4 \\
\mathbf{X}(2m, 2n) &= (\mathbf{F}_{ll}(m,n) + \mathbf{F}_{lh}(m,n) \\
&\quad + \mathbf{F}_{hl}(m,n) + \mathbf{F}_{hh}(m,n))/4.
\end{aligned} \tag{20}$$

After that, to further adjust the spectral and spatial characteristics of the fused image, the convolution block is used to refine the fusion result, and the final fused image $\hat{\mathbf{X}}$ is

$$\hat{\mathbf{X}} = \Gamma(\mathbf{X}). \tag{21}$$

To train DTAFN, the MSE loss function is used to calculate the error between the reconstructed image and the real image and is defined as follows:

$$\mathcal{L} = \sum_{i=1}^{n} (\mathbf{X} - \hat{\mathbf{X}})^2 \tag{22}$$

where $\mathcal{L}$ is the error value, $\mathbf{X}$ is the real image, and n is the number of samples.

III. Experiments

A. Experimental Settings

1) Experimental Datasets: To train DTAFN, three HSI datasets are used, including Xiongan dataset (http://www.hrscas.com/a/share/shujuchanpin/2019/0501/1049.html), Chikusei dataset (https://naotoyokoya.com/Download.html), and Pavia Center dataset (http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene). Xiongan dataset is produced by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, with a spectral range of 400–1000 nm and $3750 \times 1580 \times 250$ pixels. After clipping and removing bad bands, $3750 \times 1580 \times 100$ pixels are used for experiments. Chikusei dataset is captured by the Headwall Hyperspec-VNIR-C sensor, with a spectral range of 343–1018 nm and $2517 \times 2335 \times 128$ pixels, and the final image with $2000 \times 2000 \times 100$ pixels is used for the experiment. Pavia Center dataset is obtained by ROSIS sensor, the spectral range is 400–1000 nm, the image size is $512 \times 1400 \times 115$ pixels, and the final image with $512 \times 1400 \times 80$ pixels meets the experimental requirements. To train DTAFN and quantitatively evaluate the fusion results, the above datasets are used as reference images in the experiment. The spectral response function of SPOT-4 multispectral satellite is used to simulate MSI. In addition, Gaussian blur operation and a downsampling factor of 4 are used to simulate HSI. For the above datasets, we cut the $600 \times 600 \times 100$ pixels in the Xiongan dataset, the $600 \times 600 \times 100$ pixels in the Chikusei dataset, and the $500 \times 300 \times 80$ pixels in the Pavia Center dataset, and these data are used as test sample to verify the fusion performance of DTAFN. The size of the reference training sample, reference verification sample, and the reference test sample in the experiment is 256×256 pixels. To expand the number of samples, a spatial step of 80 is used in the image clipping process, which ensures that the number of samples in the experiment is greater than 8000.

2) Noise Setting: HSI is often contaminated with noise due to weather factors and sensor equipment, including Gaussian noise, impulse noise, strip noise, and bad lines. Moreover, many kinds of noise usually exist in HSI simultaneously, which increases the challenge of image fusion. To fully verify the anti-noise fusion performance of DTAFN, we simulated HSI with different types of mixed noise.

1) *Case 1 (Gaussian Noise + Impulse Noise):* The Xiongan dataset is used, and zero-mean Gaussian and impulse noise are added to all bands of the dataset. During the simulation, the added intensity of Gaussian noise is set to [5–30 dB] and the impulse noise is set to [0.05–0.15].
2) *Case 2 (Gaussian Noise + Impulse Noise + Strip Noise):* Chikusei dataset is used, and the same settings of Gaussian noise and impulse noise are used as in Case 1. In addition, strip noise is added to randomly selected row pixels of [5%–30%] in randomly selected 30% bands.
3) *Case 3 (Gaussian Noise + Impulse Noise + Strip Noise + Bad Lines):* Chikusei dataset is used, and the same settings of Gaussian noise, impulse noise, and strip noise are used as in Case 2. In addition, bad line is added to randomly selected row pixels of [5%–30%] in a randomly selected 30% band.
4) *Case 4 (Impulse Noise + Strip Noise):* The Xiongan dataset is used and zero-mean impulse noise is added to all bands of the dataset. In addition, strip noise is added to randomly selected row pixels in 30% of randomly selected bands [5%–30%].
5) *Case 5 (Impulse Noise + Strip Noise + Bad Line):* Pavia Center dataset is used, and the settings of impulse noise and strip noise are the same as Case 2, and the setting of bad line is the same as Case 3.
6) *Case 6 (Strip Noise + Bad Line):* Pavia Center dataset is used, and the strip noise setting is the same as Case 2 and the bad line setting is the same as Case 3.

3) Implementation and Model Training: In the experiments, we randomly selected 20% of the samples for network validation and 80% of the samples for network training. DTAFN has a learning rate of 1e−4 and uses Adam with a $\beta_1 = 0.9$ and $\beta_2 = 0.99$ as the optimizer, the training batch is set to 4, and the epoch is set to 50. Besides, the number of channels in the subspace is set to 1/4 of the number of original channels, which is validated in the discussion section. Finally, DTAFN is trained on a Window10 X64

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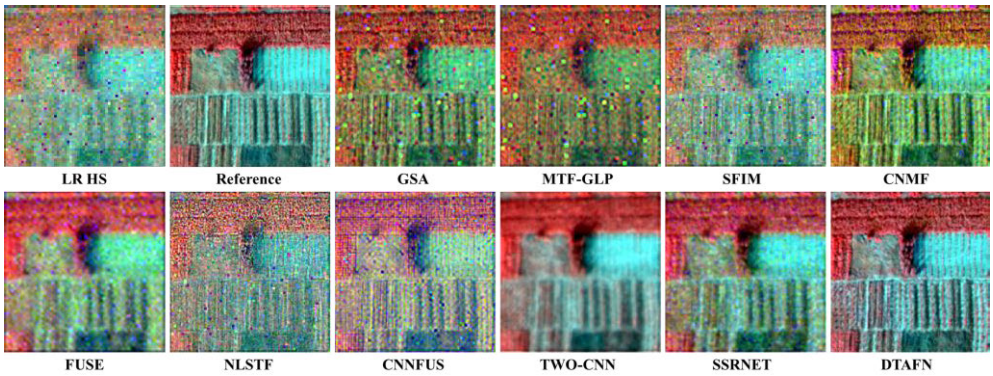


Fig. 8. Fusion results in Case 1.

computer with 64-GB CPU memory and 1× GPU of GeForce GTX 3060Ti.

4) Compared Algorithms: To evaluate the fusion performance of DTAFN, nine different traditional and deep learning methods are selected, including one CS method (GSA [15]), two MRA methods (MTF-GLP [19] and SFIM [20]), one unmixing-based method (CNMF [24]), one Bayesian-based method (FUSE [29]), one tensor decomposition-based method (NLSTF [25]), and three deep learning methods (CNNFUS [45], TWO-CNN [46], and spatial–spectral reconstruction network (SSRNET) [47]). To compare these methods fairly, the same training strategy is used, the epoch of all methods is set to 50, and the parameters are derived from relevant references.

5) Evaluation Indices: The peak signal-to-noise ratio (PSNR) [48], spectral angle map (SAM) [49], Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS) [50], root mean square error (RMSE) [51], and cross correlation (CC) [52] are used as quantitative evaluation metrics for the experiments.

1) PSNR

$$\mathrm{PSNR}(\mathbf{R}_i, \mathbf{F}_i) = 10 \cdot \log_{10}\left(\frac{\max(\mathbf{R}_i)^2}{\|\mathbf{R}_i - \mathbf{F}_i\|_2^2 / P}\right) \quad (23)$$

where $\mathbf{R}$ is the reference image, $\mathbf{F}$ is the fused image, i is the ith band, and P is the number of pixels. Higher PSNR implies better spectral and spatial fidelity.

2) SAM

$$\mathrm{SAM}\left(\mathbf{R}_j, \mathbf{F}_j\right) = \arccos\left(\frac{\mathbf{R}_j^T \mathbf{F}_j}{\left\|\mathbf{R}_j\right\|_2 \left\|\mathbf{F}_j\right\|_2}\right) \quad (24)$$

where j is the jth pixel. Higher SAM indicates more severe spectral distortion, with an ideal value of 0.

3) RMSE

$$\mathrm{RMSE}(\mathbf{R}, \mathbf{F}) = \frac{1}{PN}\|\mathbf{R} - \mathbf{F}\|_F^2 \quad (25)$$

where N is the number of bands and $\|\cdot\|_F$ is the Frobenius norm. The larger the RMSE, the greater the image reconstruction error and the worse the fusion performance.

4) CC

$$\mathrm{CC}(\mathbf{R}, \mathbf{F}) = \frac{1}{N}\sum_{i=1}^{N} \mathrm{CCS}(\mathbf{R}_i, \mathbf{F}_i). \quad (26)$$

Higher PSNR implies better spectral and spatial fidelity, where CCS calculated the correlation coefficient between two single-band images and was defined as follows:

$$\mathrm{CCS}(\mathbf{R}_i, \mathbf{F}_i) = \frac{\sum_{j=1}^{P}(\mathbf{R}_i - \mu(\mathbf{R}_i))(\mathbf{F}_i - \mu(\mathbf{F}_i))}{\sqrt{\sum_{j=1}^{P}(\mathbf{R}_i - \mu(\mathbf{R}_i))^2 \sum_{j=1}^{P}(\mathbf{F}_i - \mu(\mathbf{F}_i))^2}}. \quad (27)$$

B. Experimental Results

1) Qualitative Evaluation Results: Fig. 8 shows the fusion results of different methods in Case 1. GSA and MTF-GLP can remove some Gaussian noise but cannot suppress impulse noise, and serious spectral and spatial distortion is caused. Although the spectral difference of SFIM is small compared to the reference image, it cannot remove Gaussian noise and impulse noise, and spatial information is lost. CNMF and FUSE eliminate some Gaussian noise and impulse noise. However, CNMF has obvious spectral differences compared to the reference image, and FUSE cannot effectively enhance spatial information. NLSTF and CNNFUS obtain poor visualization results, which can not only eliminate noise but also cause serious spectral and spatial errors. TWO-CNN and SSRNET achieve the best results in all benchmark methods; however, some spatial details are lost. The reason is that TWO-CNN and SSRNET cannot converge in a limited epoch. DTAFN can effectively suppress Gaussian noise and impulse noise, and the spectral and spatial information is closest to the reference image. Fig. 9 shows the SAM maps of different methods, and the color change of the SAM maps from blue to red indicates that the spectral distortion is becoming more severe. GSA, MTF-GLP, SFIM, NLSTF, and CNNFUS have serious global spectral distortions. CNMF, FUSE, TWO-CNN, and SSRNET have some spectral distortion at the edge of the ground object. Our method achieves the best spectral fidelity.

Fig. 10 shows the fusion results of different methods in Case 2. GSA, MTF-GLP, and NLSTF remove some Gaussian noise. However, impulse noise and strip noise cannot be removed, and there are serious spectral and spatial differences compared to the reference image. CNMF has clear spatial details and can suppress impulse noise and strip noise, but some Gaussian noise cannot be eliminated, and there is obvious spectral error. FUSE, CNNFUS, and SSRNET suppress

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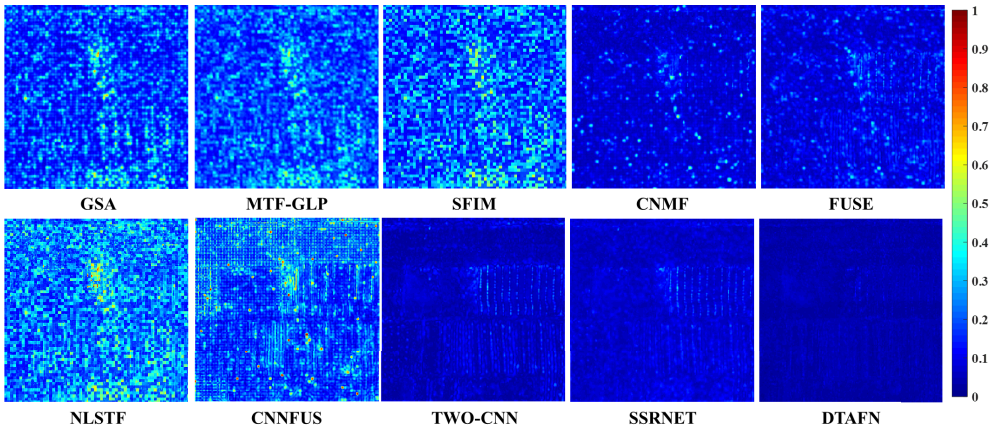


Fig. 9. SAM maps in Case 1.

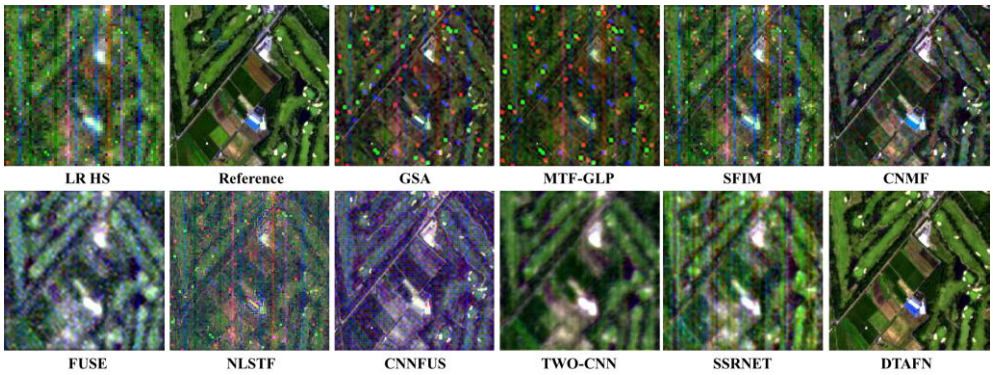


Fig. 10. Fusion results in Case 2.

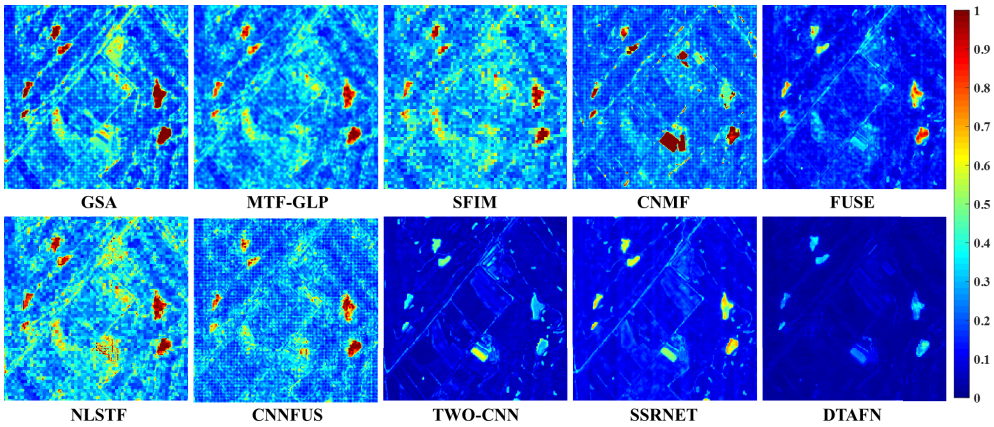


Fig. 11. SAM maps in Case 2.

impulse noise. However, FUSE cannot eliminate Gaussian noise and strip noise, and spatial and spectral information is seriously lost. CNNFUS has clear spatial details, but Gaussian noise and strip noise seriously affect fusion quality, and spectral error is obvious. SSRNET cannot eliminate the influence of Gaussian noise and strip noise, and the spatial details are seriously lost. TWO-CNN is the least affected by noise. However, spectral and spatial distortion is obvious. DTAFN obtained the closest result to the reference image. Fig. 11 shows the SAM maps of fusion results from different methods. GSA, MTF-GLP, SFIM, CNMF, NLSTF, and CNNFUS have global spectral errors, FUSE, TWO-CNN, SSRNET, and DTAFN have spectral distortions in building and water areas, and DTAFN has obtained the optimal SAM map.

The fusion result of Case 3 is shown in Fig. 12. Affected by Gaussian noise, impulse noise, stripe noise, and bad line, GSA, MTF-GLP, SFIM, FUSE, NLSTF, and CNNFUS achieve the worst fusion results. GSA, MTF-GLP, and SFIM have obvious spatial information but cannot eliminate impulse noise, strip noise, and bad line, and there is serious spectral error in the

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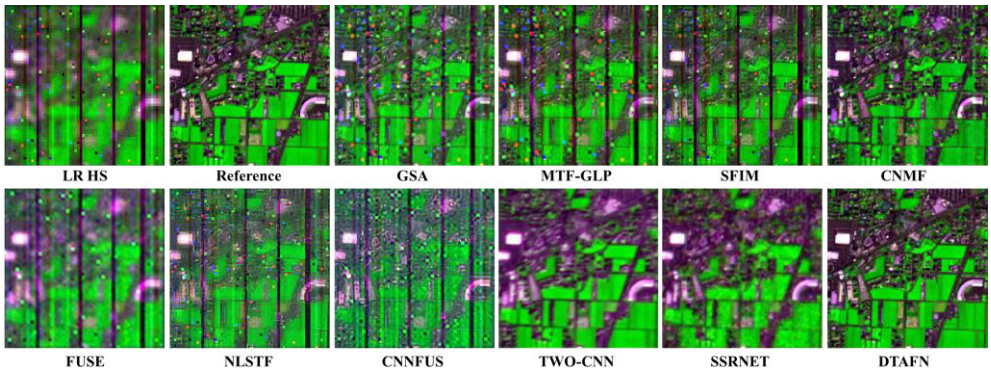


Fig. 12. Fusion results in Case 3.

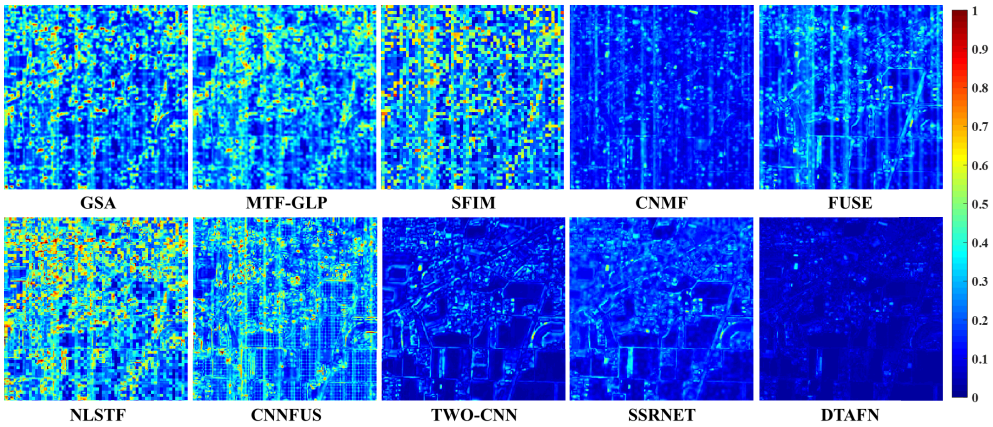


Fig. 13. SAM maps in Case 3.

building area. FUSE, NLSTF, and CNNFUS cannot eliminate Gaussian noise, impulse noise, stripe noise, and bad line, and spectral and spatial information is seriously lost. CNMF, TWO-CNN, and SSRNET have anti-noise performance. However, CNMF has spectral error in the building area, and TWO-CNN and SSRNET have lost numerous spatial information. DTAFN obtains the best visualization results. Fig. 13 shows the SAM maps of these fusion results. TWO-CNN and SSRNET have obvious spectral errors at the edge of the ground object. Other benchmark methods have global spectral errors, and DTAFN has the smallest spectral distortion.

Fig. 14 shows the fusion results of Case 4. GSA, MTF-GLP, SFIM, and NLSTF effectively enhance the spatial information of HSI but cannot eliminate impulse noise and strip noise. FUSE not only lost some spatial information but also could not suppress stripe noise. CNMF, TWO-CNN, and SSRNET have effectively suppressed impulse noise and strip noise. However, CNMF has an obvious spectral error. The spatial details of TWO-CNN are lost. SSRNET not only loses some spatial details but also has spectral error. DTAFN effectively suppresses noise and minimizes spectral and spatial errors. Fig. 15 shows the SAM maps of the fusion results. GSA, MTF-GLP, SFIM, NLSTF, and CNNFUS have serious global spectral errors. The spectral error of CNMF and FUSE mainly lies in the noise distribution position. TWO-CNN and SSRNET obtain the best spectral fidelity of all benchmark methods. Our method is superior to all benchmark methods.

The fusion result of Case 5 is shown in Fig. 16. Although GSA, MTF-GLP, and SFIM enhance the spatial resolution of the image, there are still some spatial information loss and spectral errors, and impulse noise, strip noise, and bad lines cannot be eliminated. FUSE, NLSTF, and CNNFUS can not only eliminate noise but also have obvious spectral and spatial errors. CNMF obtains the worst visualization results. TWO-CNN and SSRNET are robust for noise; however, there are spectral and spatial distortions. DTAFN has always achieved the best visualization results. In Fig. 17, TWO-CNN and SSRNET have serious spectral errors at the edge of the ground object. Other benchmark methods show poor global spectral distortion, and DTAFN has the best spectral fidelity.

Fig. 18 shows the fusion results of Case 6. Except for TWO-CNN and SSRNET, the other benchmark methods cannot suppress strip noise and bad lines. MTF-GLP, SFIM, and FUSE lose numerous spatial information, and all benchmark methods have significant spectral differences compared to the reference image. DTAFN works best in noise suppression, spectral, and spatial fidelity. Fig. 19 shows that TWO-CNN and SSRNET have severe spectral errors in the road and river, while the other benchmark methods have severe global spectral distortions. DTAFN has always had the best spectral fidelity, although there is some spectral distortion in road and river areas.

2) Quantitative Evaluation Results: Table I shows the quantitative evaluation results of different methods, and the best indexes are marked in bold. In Case 1, CNMF, FUSE,

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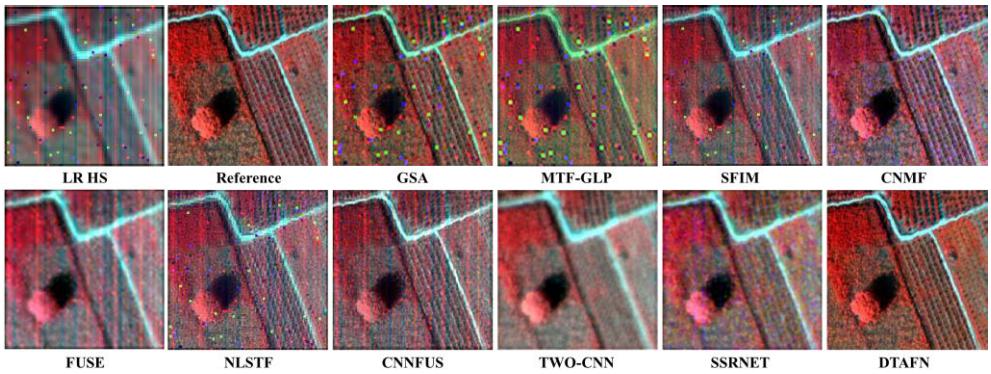


Fig. 14. Fusion results in Case 4.

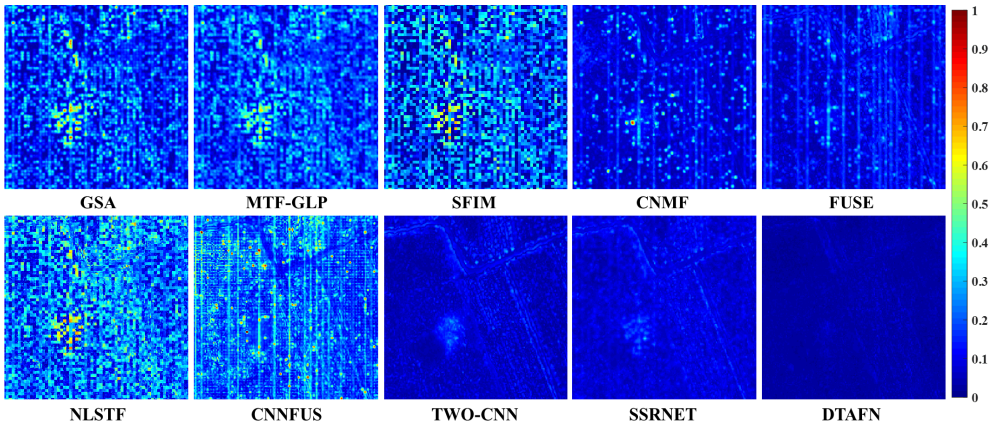


Fig. 15. SAM maps in Case 4.

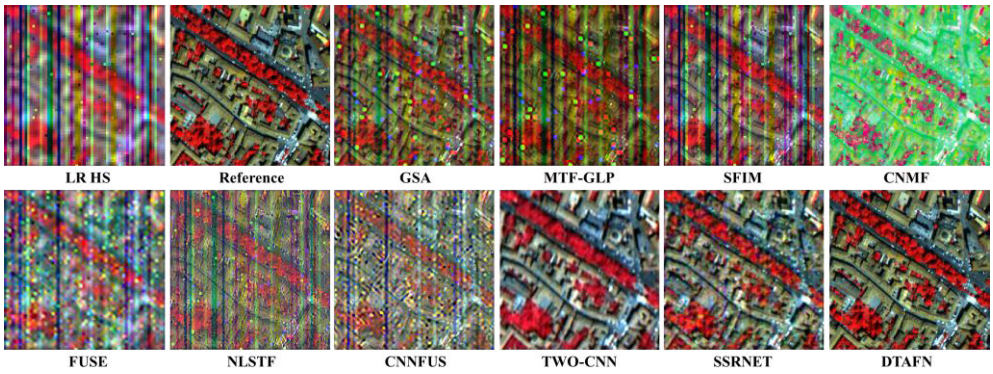


Fig. 16. Fusion results in Case 5.

TWO-CNN, and SSRNET achieve good quantitative results in all benchmark methods. However, the PSNR of these methods is less than 27, SAM and ERGAS are more than 3, RMSE is more than 0.03, and CC is less than 0.92. For Case 2, FUSE, TWO-CNN, and SSRNET achieve better evaluation results; however, PSNR, SAM, ERGAS, RMSE, and CC are lower than the evaluation results in Case 1. With more severe mixing noise, in Case 3, all the benchmark methods except TWO-CNN and SSRNET obtain poor evaluation results. When there is no Gaussian noise and impulse noise, CNMF, FUSE, TWO-CNN, and SSRNET still maintain better quantitative evaluation results among all benchmark methods, which is shown in Case 4. In Case 5, TWO-CNN obtains poor evaluation results and SSRNET still has the advantage. All benchmark methods achieve poor evaluation results in Case 6, and SAM is greater than 7, ERGAS is greater than 9, and CC is less than 0.91. Notably, DTAFN achieves the best quantitative evaluation results in all datasets, and PSNR greater than 34, SAM and ERGAS less than 3.5, RMSE is less than 0.03, and CC is greater than 0.94.

Fig. 20 shows the PSNR evaluation results for each band of all methods. PSNR fluctuates significantly between the bands of CNMF and GSA, and PSNR for the other benchmark methods is less than 30 for almost all bands. DTAFN outperforms benchmark methods in almost all bands and has a PSNR greater than 30.

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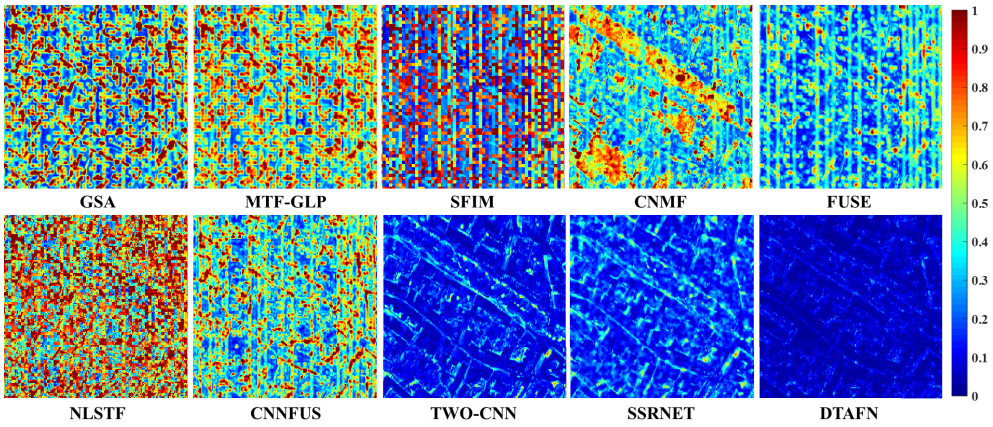


Fig. 17. SAM maps in Case 5.

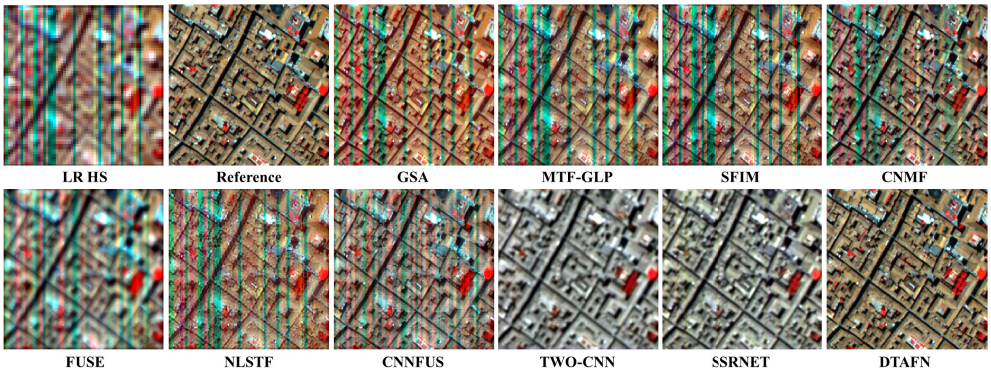


Fig. 18. Fusion results in Case 6.

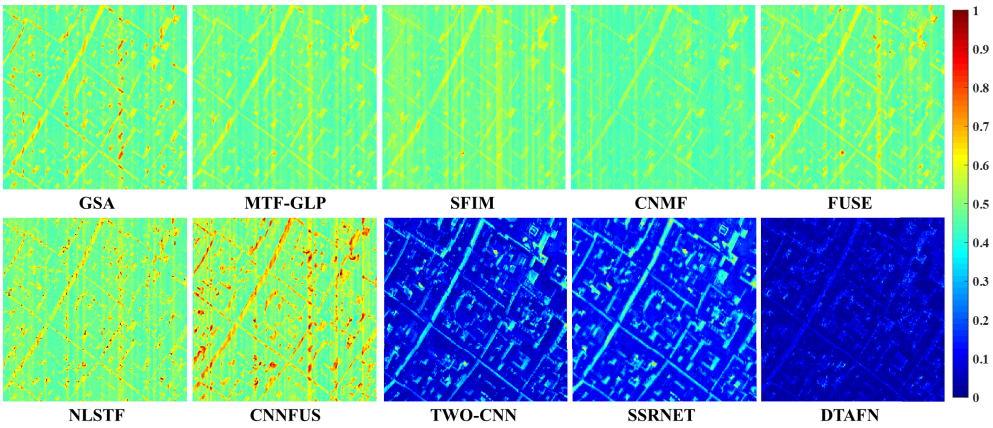


Fig. 19. Fusion results in Case 6.

C. Discussion

DTAFN combines traditional DWT and deep learning algorithms to achieve comparative fusion performance in different types of mixed noise. We believe that the main reasons are as follows: 1) DWT cannot reduce the unequal characteristics of information when enhancing the spatial resolution of HSIs, and the wrong fusion rules cause poor fusion performance. However, DTAFN uses the deep learning algorithm as the fusion rule, which solves the information asymmetry nonlinearly and drives DWT to achieve the best fusion result. 2) Considering that Conv-LSTM can reduce redundancy between hierarchical features, DTAFN uses Conv-LSTM to extract the subspaces of noise frequency components, which can effectively separate noise signals. In addition, the subspace back projection network is proposed to restore clean frequency components, which takes into account the global and local relationship of features and has more accurate reconstruction performance. 3) In DTAFN, DWT is used for frequency decomposition, which reduces the size of the original image by 1/2 and effectively improves the network operation efficiency and convergence rate.

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TABLE I
QUANTITATIVE RESULTS

Methods	GSA	MTF-GLP	SFIM	CNMF	FUSE	NLSTF	CNNFUS	TWO-CNN	SSRNET	CWTN
Case 1: Gaussian noise+impulse noise										
PSNR	17.70	16.63	15.66	26.28	21.89	14.75	18.32	19.91	22.28	**34.98**
SAM	10.97	11.94	12.69	4.93	5.68	14.57	12.31	3.25	4.14	**2.55**
ERGAS	8.48	9.13	10.42	5.05	4.85	11.39	10.39	4.19	3.44	**2.39**
RMSE	0.05	0.06	0.07	0.03	0.04	0.08	0.08	0.05	0.04	**0.02**
CC	0.71	0.60	0.57	0.90	0.79	0.54	0.65	0.91	0.90	**0.94**
Case 2: Gaussian noise+impulse noise+stripe noise										
PSNR	18.91	18.12	17.08	23.66	23.63	16.39	21.75	24.68	24.84	**34.92**
SAM	16.54	16.64	18.08	14.69	8.97	19.17	15.85	6.49	8.79	**2.10**
ERGAS	24.27	18.46	21.88	11.37	7.95	24.22	15.05	7.45	8.07	**3.15**
RMSE	0.09	0.09	0.11	0.12	0.06	0.11	0.09	0.05	0.05	**0.02**
CC	0.72	0.59	0.55	0.86	0.81	0.53	0.79	0.91	0.91	**0.99**
Case 3: Gaussian noise+impulse noise+stripe noise+bad line										
PSNR	22.87	21.55	20.64	29.43	22.90	19.53	24.09	26.13	21.06	**35.13**
SAM	16.21	17.23	17.46	8.55	11.19	20.33	16.42	4.39	7.01	**3.10**
ERGAS	10.46	11.80	13.40	7.93	9.72	15.54	11.31	7.66	14.75	**3.08**
RMSE	0.06	0.07	0.08	0.04	0.07	0.09	0.07	0.04	0.09	**0.02**
CC	0.84	0.78	0.75	0.95	0.83	0.70	0.86	0.88	0.82	**0.99**
Case 4: Impulse noise+stripe noise										
PSNR	19.87	18.58	17.71	26.80	22.07	16.53	19.84	22.17	24.60	**36.57**
SAM	10.08	11.05	10.75	5.93	6.37	13.77	13.12	3.83	4.70	**1.72**
ERGAS	9.29	9.91	11.32	7.32	6.73	13.63	12.52	4.17	3.46	**1.33**
RMSE	0.05	0.06	0.07	0.03	0.05	0.08	0.08	0.05	0.03	**0.01**
CC	0.73	0.61	0.58	0.84	0.72	0.54	0.67	0.87	0.87	**0.98**
Case 5: Impulse noise+stripe noise+bad line										
PSNR	21.82	20.27	20.20	22.56	19.04	19.57	20.20	17.77	24.45	**37.69**
SAM	28.07	27.68	28.07	26.71	27.88	28.84	30.52	7.23	9.69	**3.07**
ERGAS	11.46	11.72	12.15	10.69	12.11	12.55	12.71	10.30	6.42	**2.15**
RMSE	0.15	0.15	0.15	0.14	0.15	0.16	0.14	0.11	0.05	**0.01**
CC	0.87	0.81	0.80	0.91	0.72	0.77	0.83	0.89	0.91	**0.98**
Case 6: Stripe noise+bad line										
PSNR	22.72	21.73	20.62	28.88	25.80	18.95	26.52	17.07	25.96	**40.37**
SAM	30.83	31.58	28.90	25.90	19.83	38.13	27.46	7.84	10.51	**3.59**
ERGAS	18.50	20.08	23.29	30.43	13.26	28.92	16.22	15.49	9.06	**2.40**
RMSE	0.06	0.07	0.08	0.06	0.05	0.10	0.05	0.12	0.04	**0.01**
CC	0.62	0.43	0.40	0.79	0.56	0.36	0.68	0.87	0.90	**0.99**

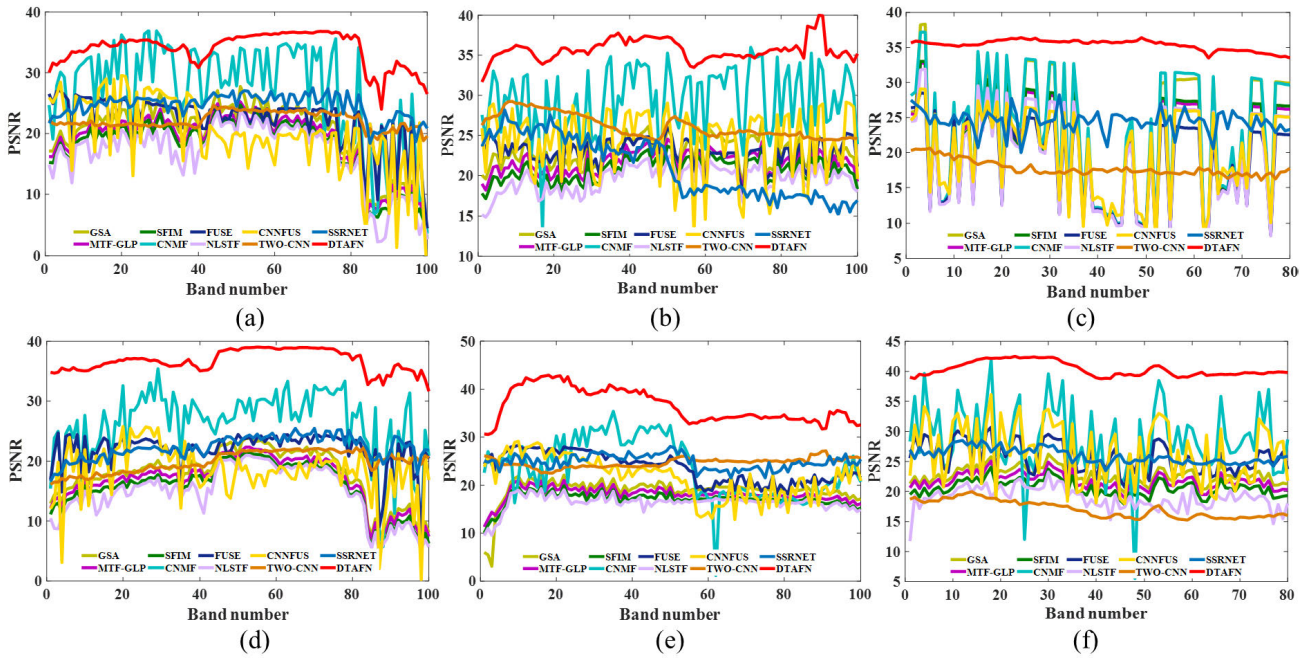


Fig. 20. PSNR of each band of fusion results in different Cases. (a) Case 1, (b) Case 2, (c) Case 3, (d) Case 4, (e) Case 5, and (f) Case 6.

To verify the effectiveness of DWT in DTAFN, we used Case 3 dataset to perform ablation experiments on the network structure. Fig. 21 shows the fusion results using only DWT, feature fusion network only, and DTAFN. Experimental results

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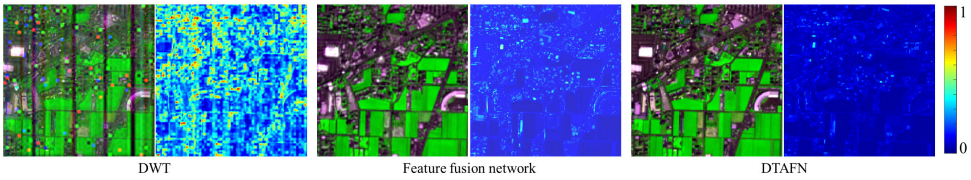


Fig. 21. Network structure ablation experimental results.

TABLE II
CALCULATION EFFICIENCY EVALUATION RESULTS

Methods	Train (min)	Test (min)
DWT	—	0.4728
Feature fusion network	237.4471	0.1506
DTAFN	193.0256	0.0833

TABLE III
NETWORK STRUCTURE ABLATION EXPERIMENT

Feature interaction	Conv-LSTM	SBPM	PSNR	SAM
√	×	×	23.98	9.62
√	√	×	28.10	6.81
√	√	√	35.13	3.10

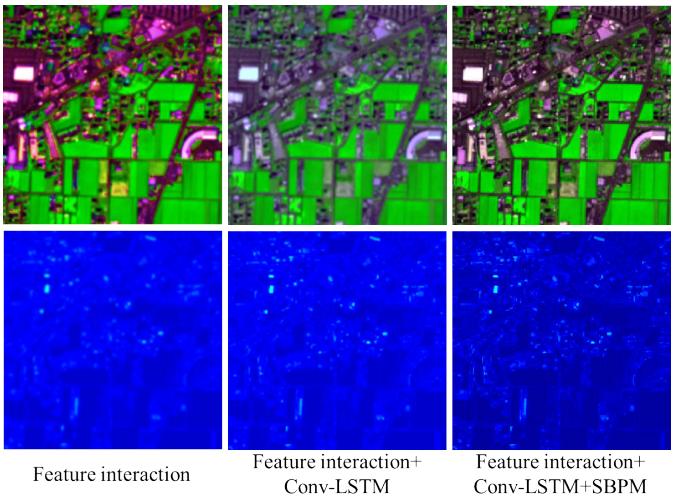


Fig. 22. Fusion results of network structure ablation experiments.

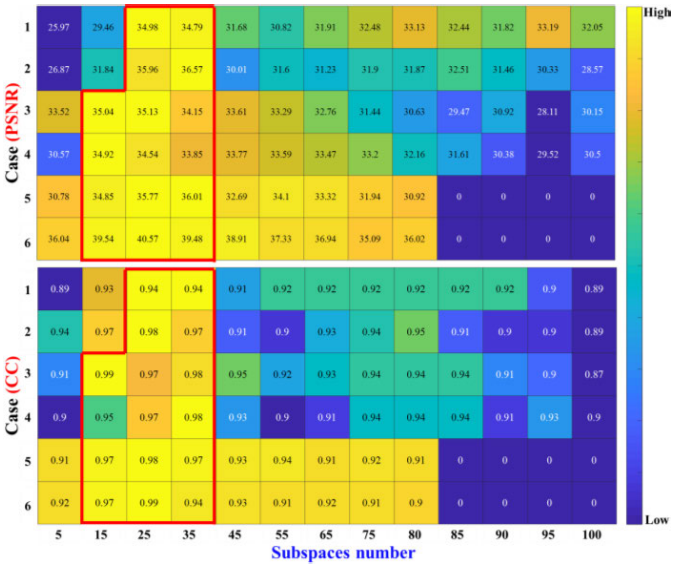


Fig. 23. Fusion results for different numbers of subspaces.

show that DWT cannot effectively suppress noise, and there is serious spectral distortion in the building area and global spatial distortion. The feature fusion network proposed by us can effectively suppress noise signals, but some spatial information is lost in the limited epoch, and spectral distortion exists in the building area. DTAFN effectively improves the spectral and spatial fidelity of fusion results. In addition, Table II shows the calculation time, and the feature fusion network effectively improves the calculation efficiency after combining with DWT.

To validate the rationality of network design, we performed ablation experiments on the network structure, including feature interaction networks without Conv-LSTM, feature interaction networks with Conv-LSTM, and feature interaction networks with Conv-LSTM combined with SBPM. Fig. 22 and Table III show the experimental results and clarify that the best fusion results are achieved when all modules are combined. While using only the feature interaction network can remove noise, it leads to significant spectral and spatial distortion.

To determine the optimal number of subspace features, we validated DTAFN. Fig. 23 shows the PSNR and CC for different numbers of subspace features. Experimental results show that when the number of subspace features is less than 5, DTAFN cannot achieve better fusion results in different Cases, and when the number is between 15 and 35, DTAFN has the best fusion performance in all cases, and fusion performance decreases as the number of features increases. Therefore, in the experiment, we set the number of subspace features to 1/4 of the number of original HSI bands.

We also compare the computational efficiency of TWO-CNN, SSRNET, and DTAFN by using the Case3 dataset in Table IV, and the experimental results show that DTAFN has higher computational efficiency.

In the end, we verified the denoising performance of DTAFN on real datasets, which were obtained from ZI-1E HSI and MSI. The HSI had a spatial resolution of 30 m and a spectral range of 400–2500 nm, with a total of 100 bands

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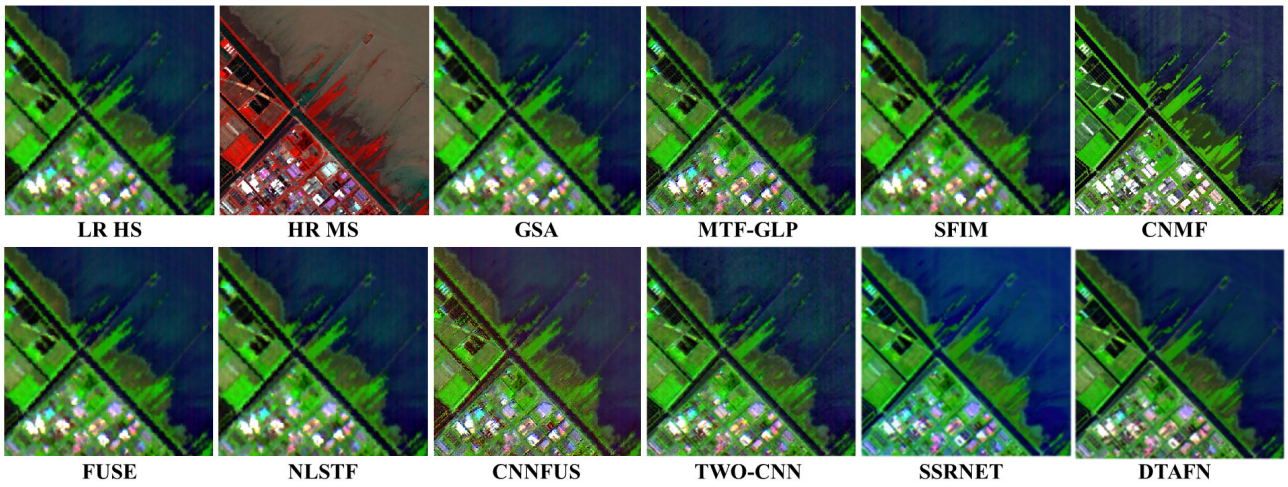


Fig. 24. Fusion results for different numbers in real dataset (band 100, band 99, and band 1).

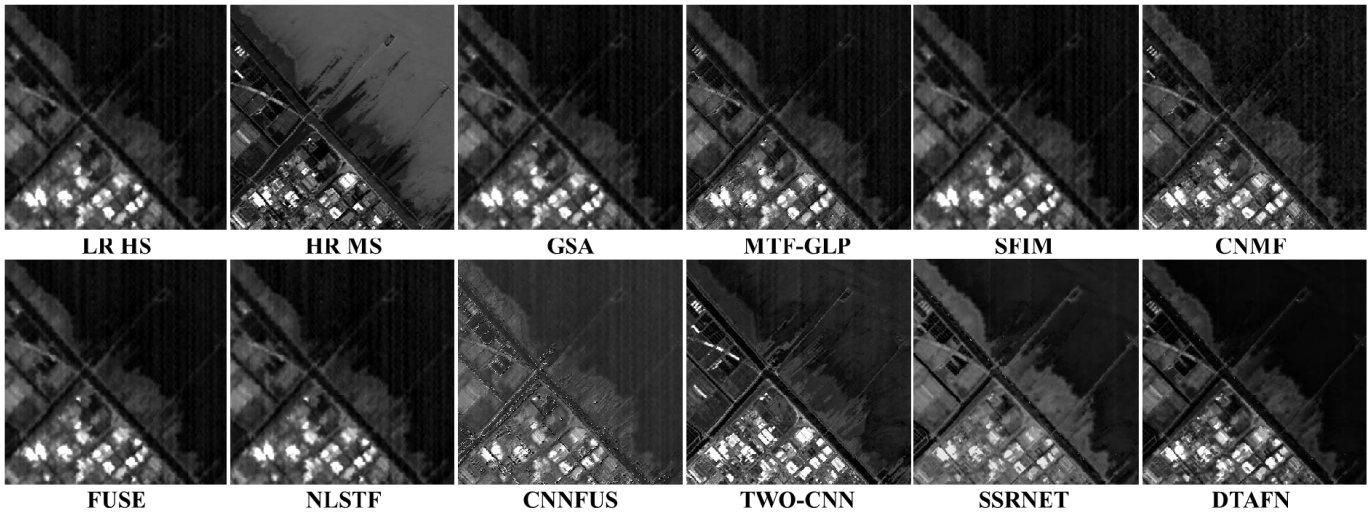


Fig. 25. Fusion results for different numbers in band 100.

TABLE IV
CALCULATION EFFICIENCY EVALUATION RESULTS OF DIFFERENT METHODS

Methods	Train (min)	Test (min)
TWO-CNN	260.5134	0.1666
SSRNET	247.0514	0.2064
DTAFN	193.0256	0.0833

used for experimentation after removing bad bands. The MSI had a spatial resolution of 10 m and included eight bands. The training dataset consisted of HSI and MSI acquired in Yancheng, Jiangsu on July 10, 2021, with the original HSI used as ground truth. The HSI was downsampled to simulate the new HSI, and the MSI was downsampled to simulate the new MSI. This HSI and MSI datasets had minimal noise and high data quality. Additionally, HSI and MSI acquired on December 2, 2021, in Yancheng, Jiangsu were used as the test dataset, which contained Gaussian noise and striping artifacts. Fig. 24 shows the visual results of different methods, illustrating that GSA, SFIM, FUSE, and NLSTF lose some spatial information during the fusion process. MTF-GLP and CNMF cause oversharpening of object edges, CNNFUS cannot effectively remove some noise, SSRNET exhibits spectral distortion, while TWO-CNN and DTAFN achieve the best visual results. Fig. 25 displays the fusion results for band 100, where GSA, MTF-GLP, SFIM, CNMF, FUSE, NLSTF, CNNFUS, and TWO-CNN fail to effectively remove striping noise. Although SSRNET removes noise, spectral distortion persists. DTAFN obtains the best denoising fusion results.

IV. CONCLUSION

Considering that the frequency domain is more suitable for describing the distribution properties of noise signals, we propose a domain transform model driven by deep learning

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for anti-noise HSI and multispectral image fusion. DTAFN first maps HSI and MSI into the frequency domain. Conv-LSTM is then used to fuse the multifrequency components of HSI and MSI and to extract the subspaces of the fused components. After that, a subspace back projection network is proposed to obtain clean frequency components with high spatial resolution. Finally, IWT and convolution layer are used to realize image fusion by inverse transformation of frequency components. Numerous mixed noise datasets are used to validate the fusion performance of DTAFN, and we make the following conclusions. 1) DTAFN can effectively fuse HSI contaminated with different types of mixed noise and MSI and has robustness. 2) DTAFN has a comparative fusion performance compared to CS, MRA, matrix decomposition, and current deep learning methods. 3) DTAFN is trained nonlinearly in low-dimensional features of images and has high computational efficiency. 4) To better train DTAFN, the number of channels in 1/4 of the original image should be set as the number of subspace features.

While DTAFN has achieved promising fusion results, it requires a substantial amount of data and memory space for network training. An effective solution is to develop an unsupervised loss function tailored to noise signal constraints, such as low-rank and sparse priors. In the future, we plan to integrate matrix decomposition models with deep learning models, using deep learning to optimize the parameters of mathematical models. This approach aims to increase the interpretability of deep learning models while enhancing the optimization and solving capabilities of mathematical models.

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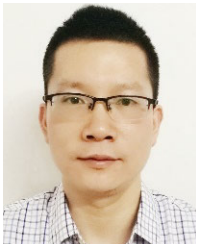
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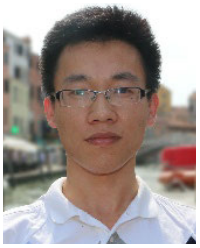
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